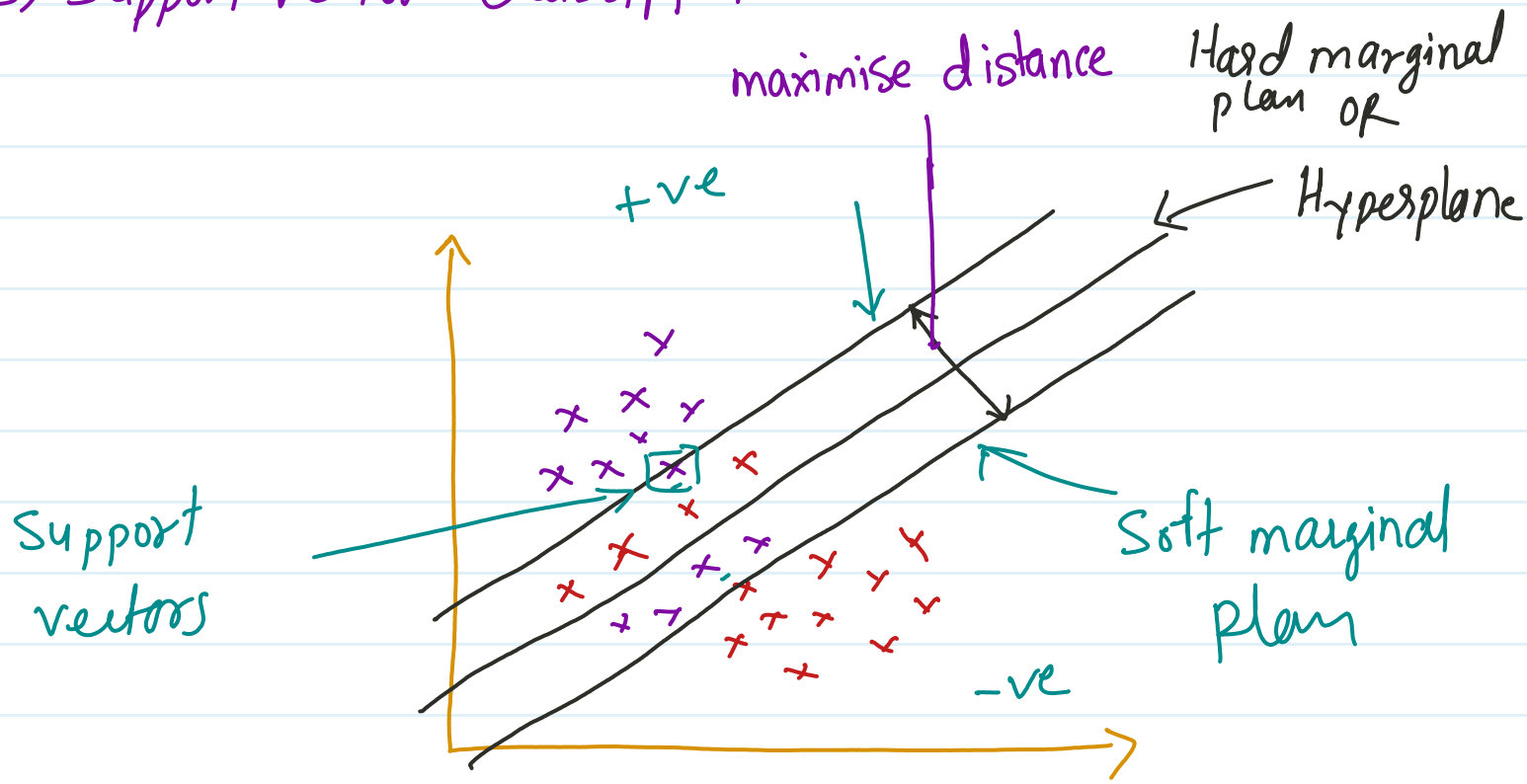
✯ Support vector machine (SVM)

SVM

- Support vector classifier
- Support vector Regressor

⇒ Support vector Classifier



$$y = mx + c$$

or

$$y = \theta_0 + \theta_1 x$$

or

$$y = \beta_0 + \beta_1 x$$

Note:- we alway play with soft marginal plan for seperate point

or

$$y = w_1 x + b$$

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

$$w^T = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

Transpose

$$y = w^T x + b$$

☆

$$ax + by + c = 0$$

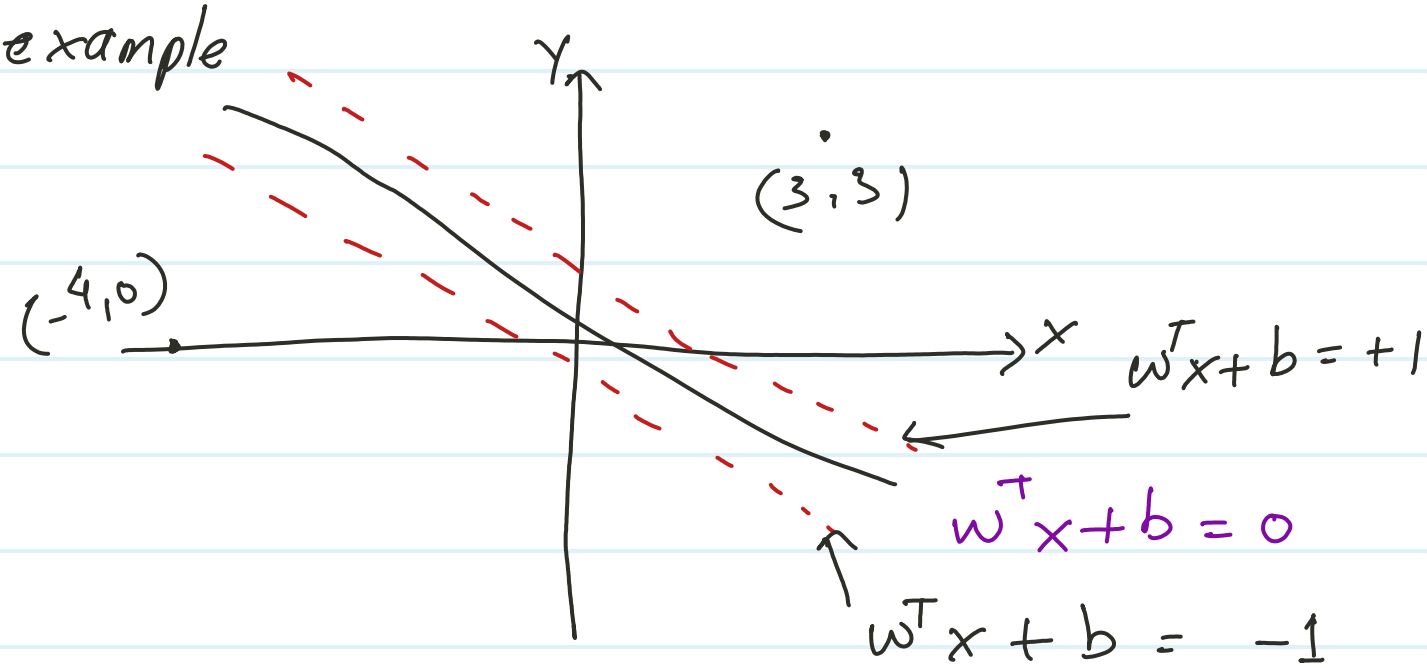
$$by = -ax - c$$

$$\boxed{y = -\frac{a}{b}(x) - \frac{c}{b}}$$

coefficient (slop) $m = -\frac{a}{b}$

Intercept $c = -\frac{c}{b}$

for example



we have a line eqn

$$3x + 2y + 4 = 0$$

for first point – (-4,0)

$\rightarrow$ $3 \times (-4) + 2 \times 0 + 4$

$\rightarrow$ $-12 + 4$

$\rightarrow$ -8 (-ve point)

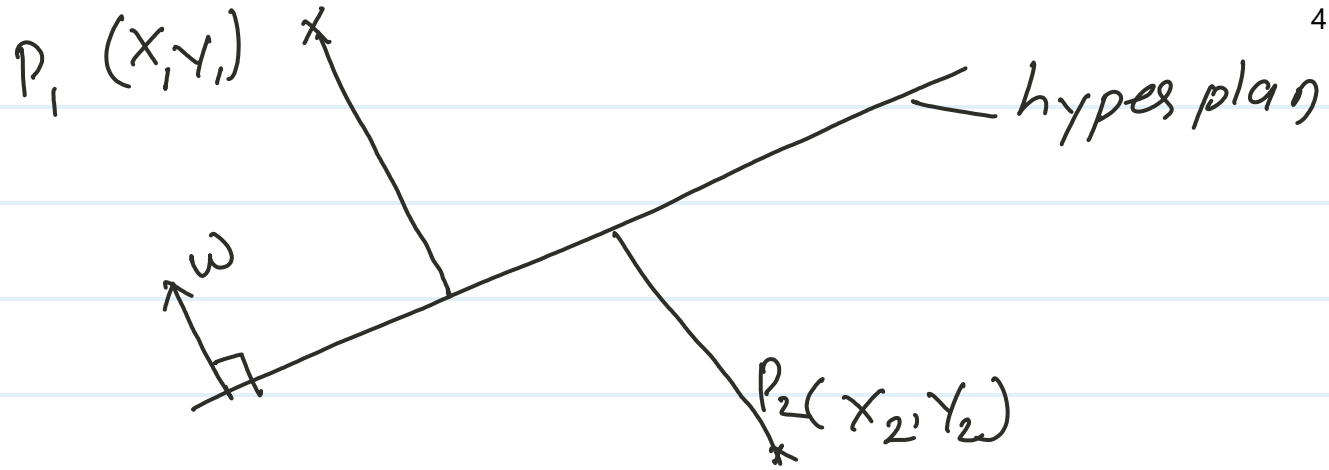
for second point – (3,3)

$\rightarrow$ $3 \times 3 + 2 \times 3 + 4$

$\rightarrow$ $9 + 6 + 4$

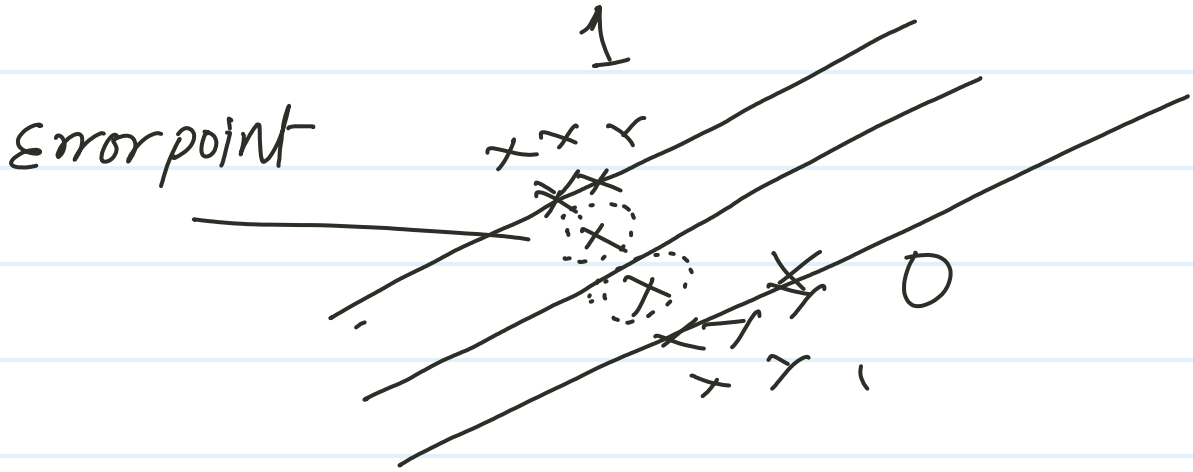
$\rightarrow$ 19 (+ve point)

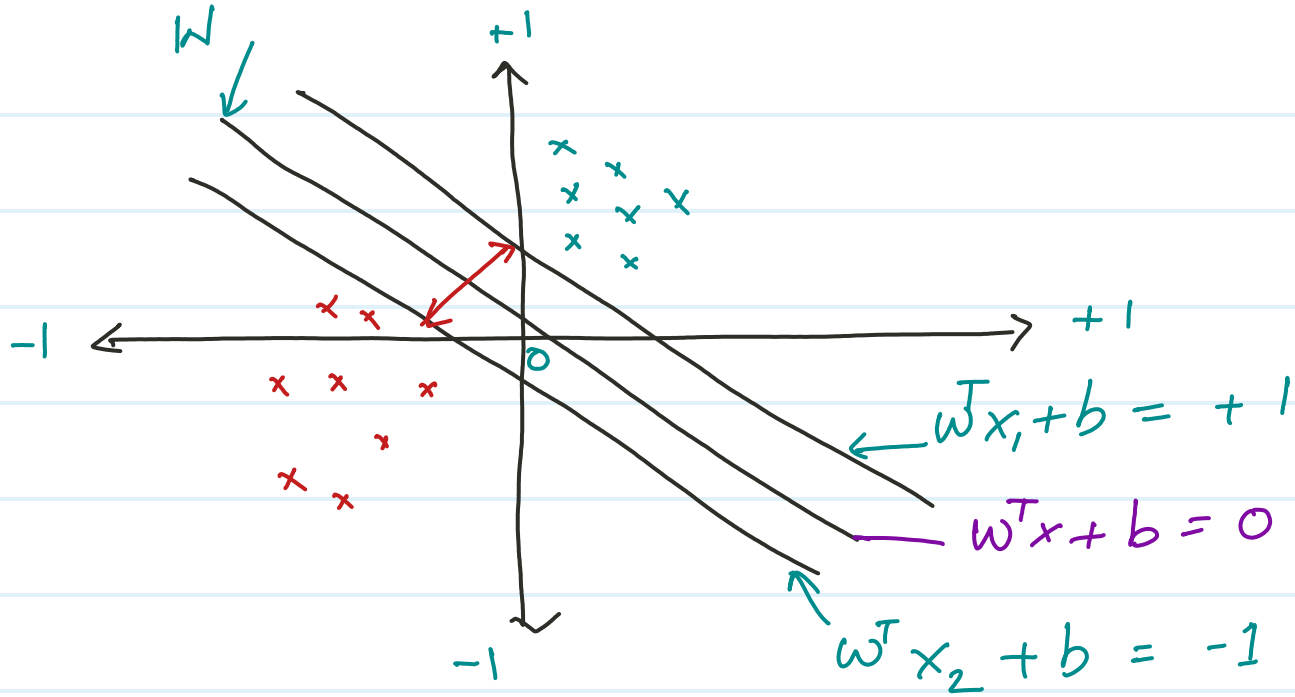
From this example if point is above line point will be positive or if point is below line point will be negative.



$$d = \frac{w^T p}{\|w\|}$$

☆ point above the line will be positive and below the line will be negative.





$$w^T x_1 + \cancel{b} = +1$$

$$- \quad w^T x_2 \pm \cancel{b} = \mp 1$$

$$w^T (x_1 - x_2) = 2$$

The eqn divied by $\|w\|$ (magnitude of w), To get unit vector.

$$\frac{w^T (x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|} \Big\} \text{ maximize}$$

constraint $y_i \begin{cases} +1, & w^T x + b \geq 1 \\ -1, & w^T x + b \leq -1 \end{cases}$

for all the correct point

$$\text{constant} = y_i \times (w^T x + b) \geq 1$$

Revers the eqn to min. magnitude

$$\boxed{\max_{(w,b)} = \frac{2}{\|w\|}} \qquad \min_{w,b} = \frac{\|w\|}{2}$$

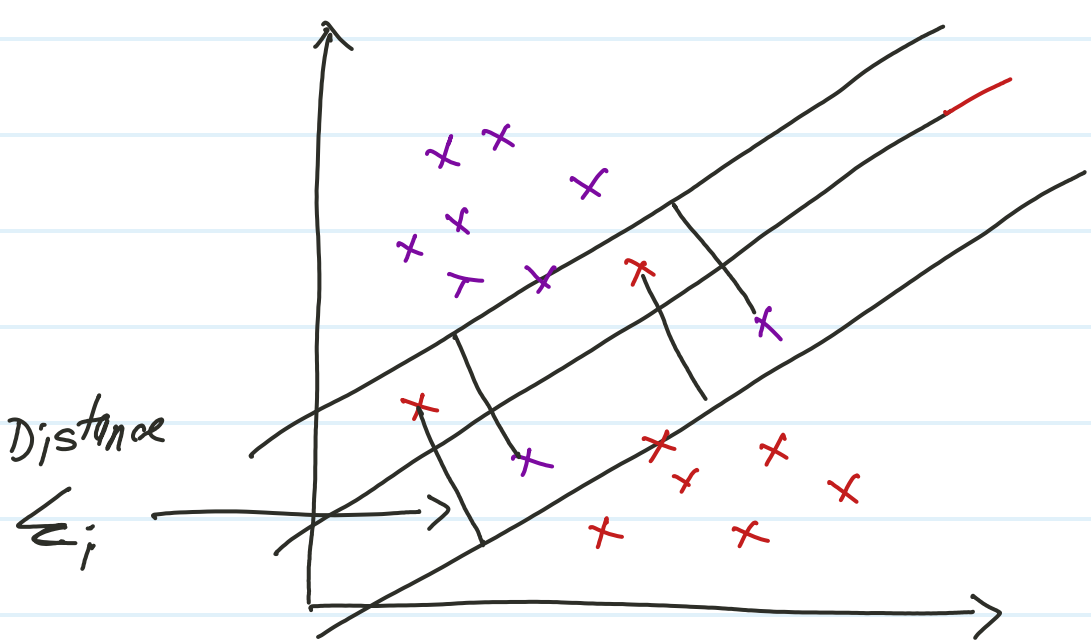
marginal plan distance

☆ Cost function

$$\min_{(w,b)} = \frac{\|w\|}{2} + \boxed{C_i \sum_{i=1}^{w} \varepsilon_i}$$

— hingeloss

To avoid point for misclassification



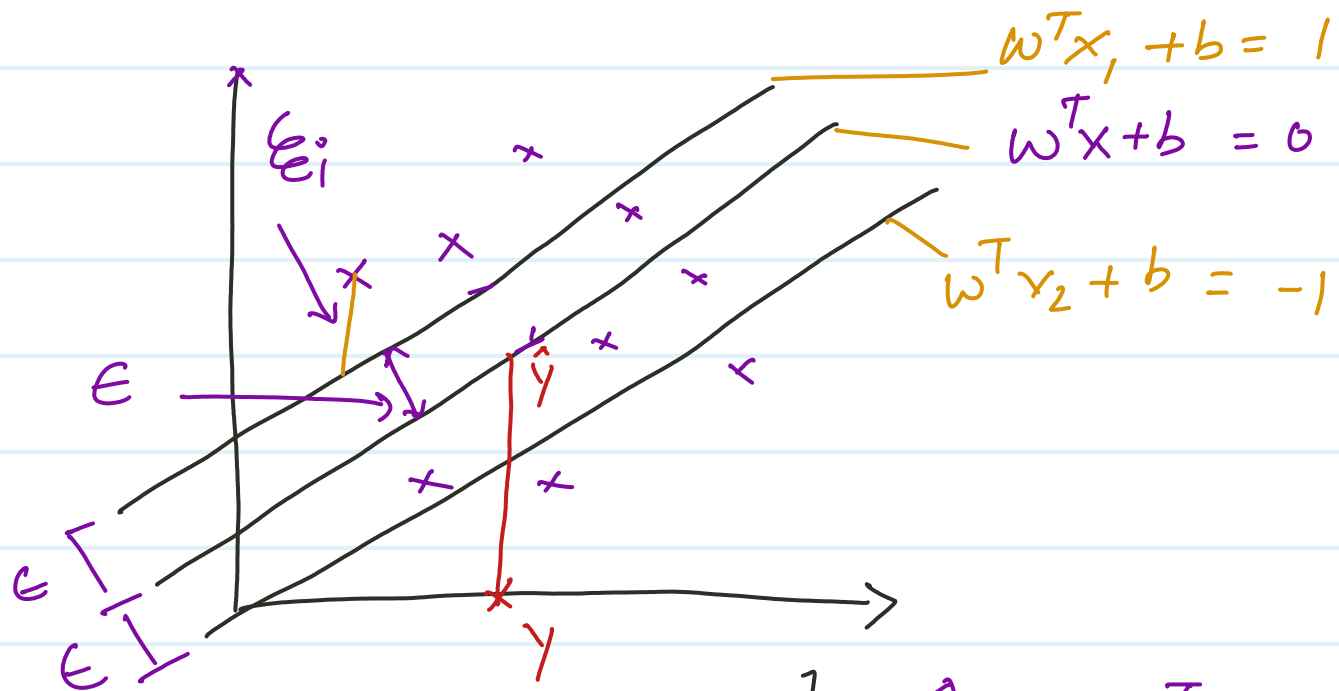
$C_i = 4$

C_i = How many point we can ignore for misclassification

ξ_i = Sum of the distance of misclassified point from marginal plan / hyperplan

Hyperparameter.

☆ SVR (Support Vector Regressor)



$w^T x + b = 0$

$w^T x - \epsilon$

$\hat{y} = w^T x_i$

cost function

$$\min_{(w,b)} \frac{\|w\|}{2} + \boxed{c_i \sum_{i=1}^{n} |\varepsilon_i|} \text{ — Hinge loss}$$

constraint $[y_i - w^T x_i] \leq \boxed{\epsilon + |\varepsilon_i|}$

hyperparameter

Limitation -

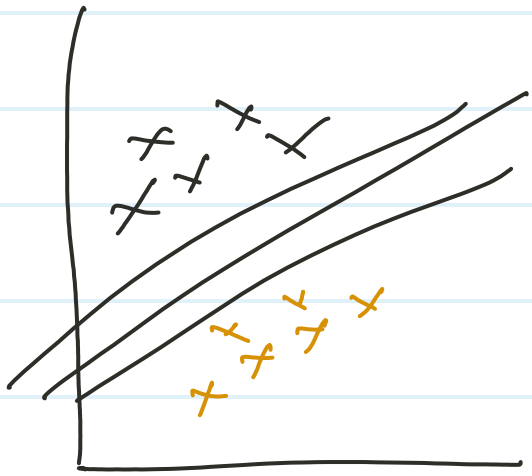
1. Impact by outlier
2. Required standarized data point x and y

Advantages -

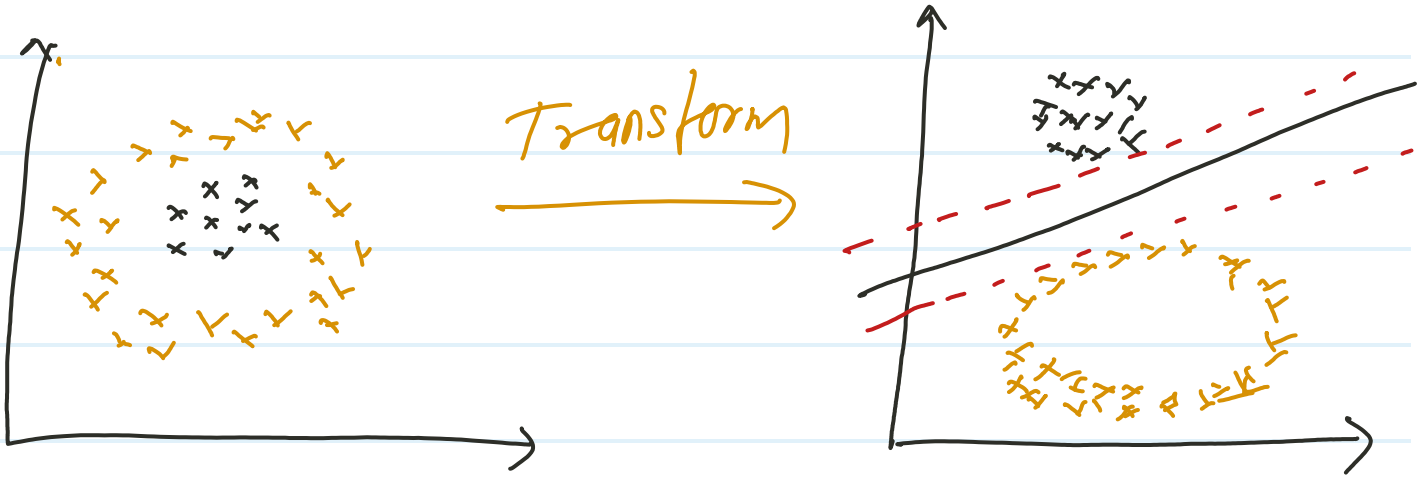
1. SVM is used solve multi class classification problem
2. Even it does not required Linear data like logistic and linear Regression for model building.

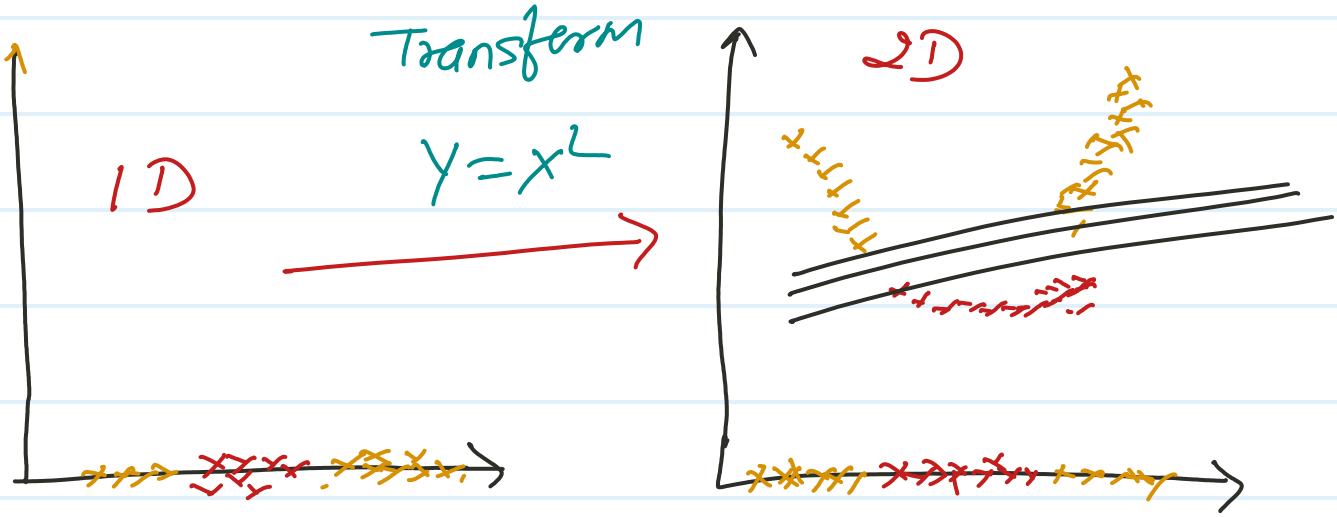
SVM kernel

SVM kernel only used for classification problem



Linear seprable





1. polynomial kernal
2. RBF kernel
3. sigmoid kernel

1. polynomial kernal

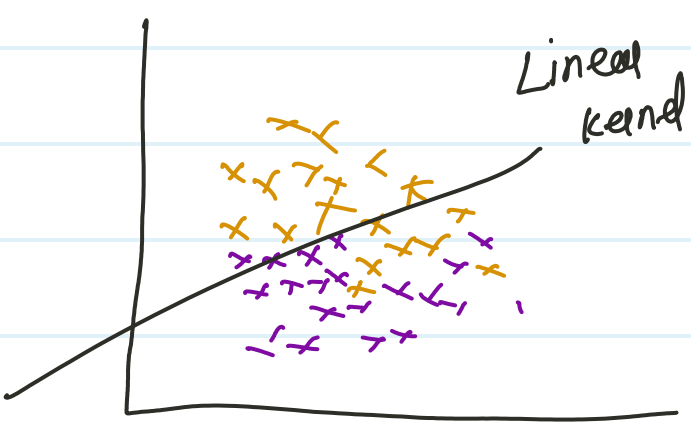
we convert Datapoint from 2D to 3D

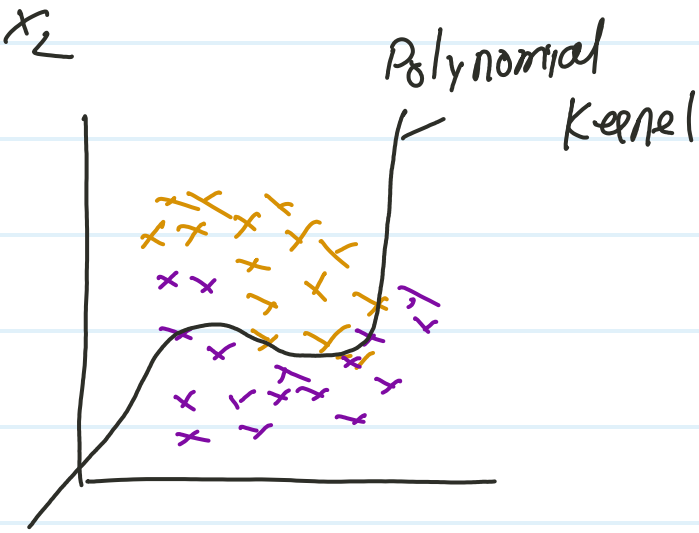
$$f(x_1, x_2) = (x_1^T x_2 + 1)^d$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1^2 & x_1 x_2 \\ x_1 x_2 & x_2^2 \end{bmatrix}$$

$x_1 \quad x_2 \qquad x_1 x_2 \qquad x_1 x_2$



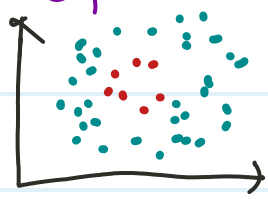


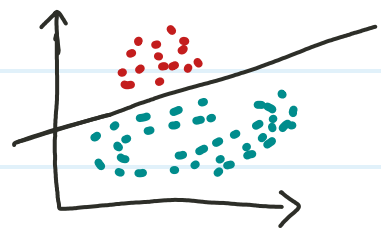
2. RBF (Radial basis kernel)

We change dimension in it.

$$k(\vec{x}, l^i) = \frac{e^{-\|\vec{x} - \bar{l}^i\|^2}}{2\sigma^2}$$

it is used to separate circular data point..





③ Sigmoid kernel ⇒

1D data point

$$\Rightarrow \frac{1}{1+e^{-z}}$$